Record Reproduction Certification

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Product and Lot Information

Product	OBX-201 (recombinant AAV9)	Drug substance lot	DS-26052
Process step	Production bioreactor, triple transfection	Manufacturing date	10JUN2026
Cell line	Suspension HEK293	Working cell bank lot	WCB-293-018
Production bioreactor ID	BRX-200-02	Working volume (mL)	10,000
Transfection reagent	Linear PEI (1 mg/mL)	Target harvest	72 h post-transfection

Referenced SOPs

SOP Number	Title
SOP-3010	Working Cell Bank Thaw and Seed Train Expansion
SOP-3040	N-1 Seed Bioreactor Operation
SOP-3110	Triple Transfection and Production Culture
SOP-3120	Production Bioreactor In-Process Monitoring
SOP-3150	Harvest and Lysis
SOP-1203	Buffer and Solution Preparation
SOP-0905	Bioburden and Endotoxin Sampling

1. EQUIPMENT / INSTRUMENTS

Description	EIN	Calibration Due	Performed By / Date
Production bioreactor skid	BRX-200-02	30SEP2026	TN 10JUN26
Dissolved oxygen probe	DO-207	12OCT2026	TN 10JUN26
pH probe	PHM-209	05NOV2026	TN 10JUN26
Viable cell counter	VCC-011	20AUG2026	TN 10JUN26
Analytical balance	BAL-021	22NOV2026	TN 10JUN26
Biosafety cabinet	BSC-06	15DEC2026	TN 10JUN26

2. PLASMID AND REAGENT PREPARATION

Component	Identity / Composition	Lot Number	Prepared By / Date	Verified By / Date
Rep/Cap plasmid	pOBX-RepCap (AAV9)	PL-REPCAP-0417	TN 10JUN26	PA 10JUN26
Helper plasmid	pHelper (Ad E2A, E4, VA RNA)	PL-HELP-0417	TN 10JUN26	PA 10JUN26
Transgene plasmid	pOBX-ITR-transgene	PL-TRANS-0417	TN 10JUN26	PA 10JUN26
Transfection reagent	Linear PEI, 1 mg/mL	PEI-0392	TN 10JUN26	PA 10JUN26
Production medium	Chemically defined, serum-free	MED-2208	TN 10JUN26	PA 10JUN26

3. SEED AND INOCULATION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
3.1	Record viable cell density at inoculation.	Viable cell density (e6 cells/mL). Acceptance: 0.5 to 1.5.	0.9	TN 10JUN26	PA 10JUN26
3.2	Record viability at inoculation.	Viability (percent). Acceptance: not less than 90.	95	TN 10JUN26	PA 10JUN26
3.3	Confirm working volume.	Working volume (mL). Acceptance: 10,000 plus or minus 500.	10,000	TN 10JUN26	PA 10JUN26

4. TRANSFECTION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
4.1	Record viable cell density at transfection.	Viable cell density (e6 cells/mL). Acceptance: 1.0 to 2.0.	1.7	TN 10JUN26	PA 10JUN26
4.2	Record culture volume at transfection.	Culture volume (mL). Acceptance: record.	10,000	TN 10JUN26	PA 10JUN26
4.3	Calculate total plasmid DNA for the transfection.	Total DNA (mg). Acceptance: 1.0 microgram/mL per SOP-3110. Formula: Total	DNA per mL (microgram/mL): 1.0 Culture volume (mL): 10,000	TN 10JUN26	PA 10JUN26

		DNA (mg) = DNA per mL (microgram/mL) x Culture volume (mL) / 1000	Result (mg): 10.0		
4.4	Calculate PEI mass from the DNA to PEI ratio.	PEI mass (mg). Acceptance: DNA to PEI 1:2 by weight per SOP-3110. Formula: PEI (mg) = Total DNA (mg) x 2	Total DNA (mg): 10.0 DNA:PEI ratio: 1:2 Result (mg): 20.0	TN 10JUN26	PA 10JUN26
4.5	Split plasmid DNA by molar ratio.	Rep/Cap : Helper : Transgene = 1:1:1 (mg each).	3.33 / 3.33 / 3.33	TN 10JUN26	PA 10JUN26
4.6	Record DNA-PEI complexation time before addition.	Complexation time (min). Acceptance: 15 to 30 min.	20	TN 10JUN26	PA 10JUN26
4.7	Confirm addition of transfection complex to the reactor.	Complex added to reactor?	☐X Yes ☐ No	TN 10JUN26	PA 10JUN26

5. PRODUCTION CULTURE

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
5.1	Record temperature setpoint.	Temperature (C). Acceptance: 37.0 plus or minus 0.5.	37.0	TN 10JUN26	PA 10JUN26
5.2	Record dissolved oxygen setpoint.	Dissolved oxygen (percent). Acceptance: 40 plus or minus 10.	40	TN 10JUN26	PA 10JUN26
5.3	Record hour 24 viability.	Viability (percent). Acceptance: record.	96	TN 10JUN26	PA 10JUN26
5.4	Record hour 24 culture pH.	pH. Acceptance: 7.0 to 7.2.	6.84	TN 10JUN26	PA 10JUN26
5.5	Record hour 48 viability.	Viability (percent). Acceptance: record.	91	TN 10JUN26	PA 10JUN26
5.6	Record hour 48 culture pH.	pH. Acceptance: 7.0 to 7.2.	7.05	TN 10JUN26	PA 10JUN26
5.7	Record hour 72	Viability (percent).	79	TN 10JUN26	PA 10JUN26

	viability.	Acceptance: record.			

6. HARVEST

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
6.1	Record harvest time post-transfection.	Harvest time (h). Acceptance: 48 to 96, target 72.	80	TN 14JUN26	PA 14JUN26
6.2	Record harvest viability.	Viability (percent). Acceptance: not less than 70.	68	TN 14JUN26	PA 14JUN26
6.3	Record harvest working volume.	Volume (mL). Acceptance: record.	9,850	TN 14JUN26	PA 14JUN26

7. IN-PROCESS SAMPLING

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
7.1	Record in-process vector genome titer (qPCR).	VG titer (e10 vg/mL). Acceptance: record.	8.9	TN 14JUN26	PA 14JUN26
7.2	Confirm bioburden sample collected.	Bioburden sample collected?	☐X Yes ☐ No	TN 14JUN26	PA 14JUN26
7.3	Confirm mycoplasma sample collected.	Mycoplasma sample collected?	☐X Yes ☐ No	TN 14JUN26	PA 14JUN26

8. PROCESS COMMENTS

Page / Step	Comment	Performed By / Date	Verified By / Date
Step 4.4	PEI mass initially calculated as 30.0 mg applying a 1:3 ratio; SOP-3110 specifies 1:2. Error identified at verification before complex preparation and recalculated to 20.0 mg. No PEI added at the incorrect mass. Ref DEV-26-0205.	TN 10JUN26	PA 10JUN26
Step 5.4	Hour 24 culture pH 6.84, below the 7.0 to 7.2 range. CO2 sparge reduced; pH recovered to 7.08 by hour 30. Ref DEV-26-0206.	TN 10JUN26	PA 10JUN26

Step 6.1/6.2	Harvest performed at 80 h against the 72 h target after a scheduling delay; harvest viability 68 percent, below the not-less-than-70 percent target. Material retained; in-process titer within expected range. Ref DEV-26-0208.	TN 14JUN26	PA 14JUN26

BATCH RECORD APPROVALS

Manufacturing Review	Quality Assurance Review
F. Castellano 16JUL2026	L. Marchetti 16JUL2026

Manufacturing and QA review signatures are applied at batch disposition.